## 1.Basic program : print “Hello World”

main.c

```c
#include <stdio.h>

int main()
{
    printf("Hello World");
    return 0;
}
```

Output

```
Hello World

=== Code Execution Successful ===
```

## Variables

## What are the rules for naming variables?

## Rules for naming variables:

All variable names must begin with a letter of the alphabet or an underscore (_).
After the first initial letter, variable names can also contain letters and numbers.
Uppercase characters are distinct from lowercase characters (case-sensitive).
You cannot use a C++ keyword (reserved word) as a variable name.

## 2.Program that represent data types:-

main.c

```c
#include <stdio.h>
int main() {
    // Integer variable
    int integerVariable = 10;
    // Floating-point variable
    float floatVariable = 3.14;
    // Character variable
    char charVariable = 'A';
    // Double-precision floating-point variable
    double doubleVariable = 2.71828;
    // Printing values
    printf("Integer: %d\n", integerVariable);
    printf("Float: %.2f\n", floatVariable);
    printf("Character: %c\n", charVariable);
    printf("Double: %.5lf\n", doubleVariable);
    return 0;
}
```

Output

```
Integer: 10
Float: 3.14
Character: A
Double: 2.71828


=== Code Execution Successful ===
```

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## Data Types

Common Data Types
Size and Range of Each Data Type

## Data Types in C

## 1. Primary (Basic Data Types)

Integer
Character
Floating Point
Double Floating Point
Void

**2. Derived Data Types**
Function
Array
Pointer
Reference

**3. User Defined Data Types**
Class
Structure
Union
Enum
Typedef

| Data Type | Range | Bytes | Format |
|---|---|---|---|
| signed char | -128 to + 127 | 1 | %c |
| unsigned char | 0 to 255 | 1 | %c |
| short signed int | -32768 to +32767 | 2 | %d |
| short unsigned int | 0 to 65535 | 2 | %u |
| signed int | -32768 to +32767 | 2 | %d |
| unsigned int | 0 to 65535 | 2 | %u |
| long signed int | -2147483648 to +2147483647 | 4 | %ld |
| long unsigned int | 0 to 4294967295 | 4 | %lu |
| float | -3.4e38 to +3.4e38 | 4 | %f |
| double | -1.7e308 to +1.7e308 | 8 | %lf |
| long double | -1.7e4932 to +1.7e4932 | 10 | %Lf |
| Note: The sizes and ranges of int, short and long are compiler dependent. Sizes in this 07:07 re are for 16-bit compiler. | | | |

**3.Program that checks if the given number is greater than the threshold value or not.**

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main.c  Share  Run

```c
#include <stdio.h>
#define MAX 100
const int MAX_VALUE = 100;
const float PI = 3.14159;
int main() {
    int inputValue;
    printf("Enter an integer value (up to %d): ", MAX_VALUE);
    scanf("%d", &inputValue);
    if (inputValue <= MAX_VALUE) {
        printf("You entered: %d\n", inputValue);
    } else {
        printf("Value exceeds the maximum limit of %d\n",
            MAX_VALUE);
    }
    printf("Value of PI is: %.5f\n", PI);
    return 0;
}
```

Output  Clear

```
Enter an integer value (up to 100): 67
You entered: 67
Value of PI is: 3.14159


=== Code Execution Successful ===
```

**Constants**

**Constant Types**

Numerical Constant
Integer Constant
Decimal
Octal
Hexadecimal
Real Constant
Character Constant
String Constant

**Arithmetic operators:-**

| Name | Operator | Example |
|---|---|---|
| Addition | + | num1 + num2 |
| Subtraction | - | initial - spent |
| Multiplication | * | fathoms * 6 |
| Division | / | sum / count |
| Modulus | % | m % n |

## 4.Program that shows all the arithmetic calculations done using arithmetic operators.

main.c

Share Run

```c
#include <stdio.h>

int main() {
    int a = 10, b = 3;

    printf("Addition: %d\n", a + b);
    printf("Subtraction: %d\n", a - b);
    printf("Multiplication: %d\n", a * b);
    printf("Division: %d\n", a / b);
    printf("Modulus: %d\n", a % b);

    return 0;
}
```

Output Clear

```
Addition: 13
Subtraction: 7
Multiplication: 30
Division: 3
Modulus: 1


=== Code Execution Successful ===
```